

Console window
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modeling in programming

OOA & PD

UML → Unified modeling language

attributes are properties of a class processed by all objects

A method is an action that can be requested from any object in the class.

- ✓ class diagrams
- ✓ Use cases
- ✓ Deployment diagrams
- ✓ Sequence diagrams
- etc.

CATEGORIES OF UML

1. Objects → created from a class
2. Relationships → This is the glue that holds things together.
3. Diagrams → This are categorized as either structural or behavioural

GROUPING IN UML

(i) Structural grouping

This defines the conceptual & physical structures of an oop system and are described by nouns.

(ii) behavioural grouping

This represents the behavior of the system & the states of the system before, during & after the behaviour action occurs.

RELATIONSHIPS IN UML

(i) Structural

This depends upon the structure; Some of the properties are

- i) dependencies

ii) Aggregations

iii) Associations

iv) Generalizations

→ UML is a modeling language for specifying, visualizing, constructing and documenting the details of any computer system-intensive processes.

→ It was conceived by Rational Software Corporation & 3 of the most prominent methodologists in the info systems & tech industry:

Grady Booch, James Rumbaugh, and Ivan Jacobson

oop concepts

The concepts are:

- instance of a class

(i) an object which represents a real world thing or event. The objects may be customers, orders and so on.

The objects may be GUI displays or text areas on a display. The objects are represented by and grouped into classes. A class or group of related objects has a set of shared attributes & behaviours

instantiation is a term used when an object is created from a class

(2) It identifies external actors

(ii) Behavioral

depends upon the behavioural properties are:

1. Communicates
2. Includes
3. Extends
4. Generalizes

(4) It describes scenarios of the system on basis of goals and use

(5) It describes pre/post conditions.

Structural Diagrams

1. Class diagrams

2. Component

3. Deployment

4. Object

5. Package

6. Profile

Behavioral Diagrams

1. Use case Diagrams

2. Sequence

3. Collaboration

4. Statechart

5. Activity

6. Timing

(6) It describes variants / exceptions

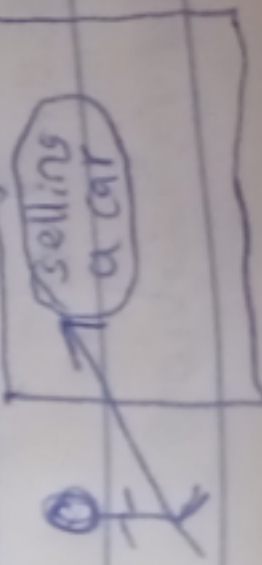
→ The use case diagrams consist of 2 parts i.e picture & description.

→ The various symbols used are:

Actor → This is a stick figure of a man: An actor / user that initiates an event.

Use Case → This is an event or operation that triggers a use case, it is represented by an oval. Selling a car

Connecting Lines → These are single lines that perform the actions triggered by the event from the actor



Use Cases

The Describe the system & what it does from the standpoint of an external observer. The various principles used in use cases are:

(1) It describes required functionality in terms of the user of the system.

(v) The arrow goes from extended to basic use case.

(x) It generalizes the use case.

The actors have the following properties:

- (1) Represents the role played by one or more users.
- (2) It exists outside of the system.

Steps for creating a use case model.

- Review the business specifications and identify the actors within the problem domain.
- Identify the high level events that develop the primary use cases that describe the events and how actors initiate them.
- Review each primary use case to determine possible variations of flow through the use case.
- Develop the use case documents for all primary use cases and all important use case scenarios.

Ex. 1 → Rent a car system is shown below.

The company does car renting or hiring, there are 2 actors as administrator & customer. The use case diagram can be prepared independently for each actor.

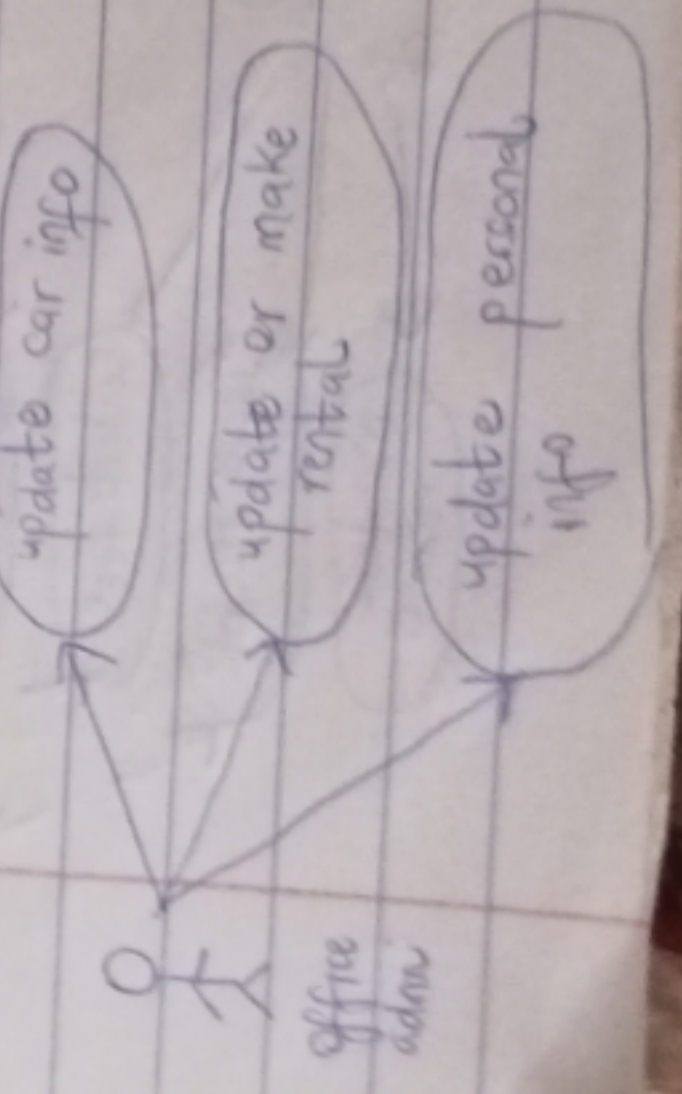
(v) The common use case is included in other use cases

(vi) The dotted arrow points towards the common use case

(vii) It 'extends' the use case

(viii) A different use case handles variations or exceptions from the basic use case.

1. Admin use case

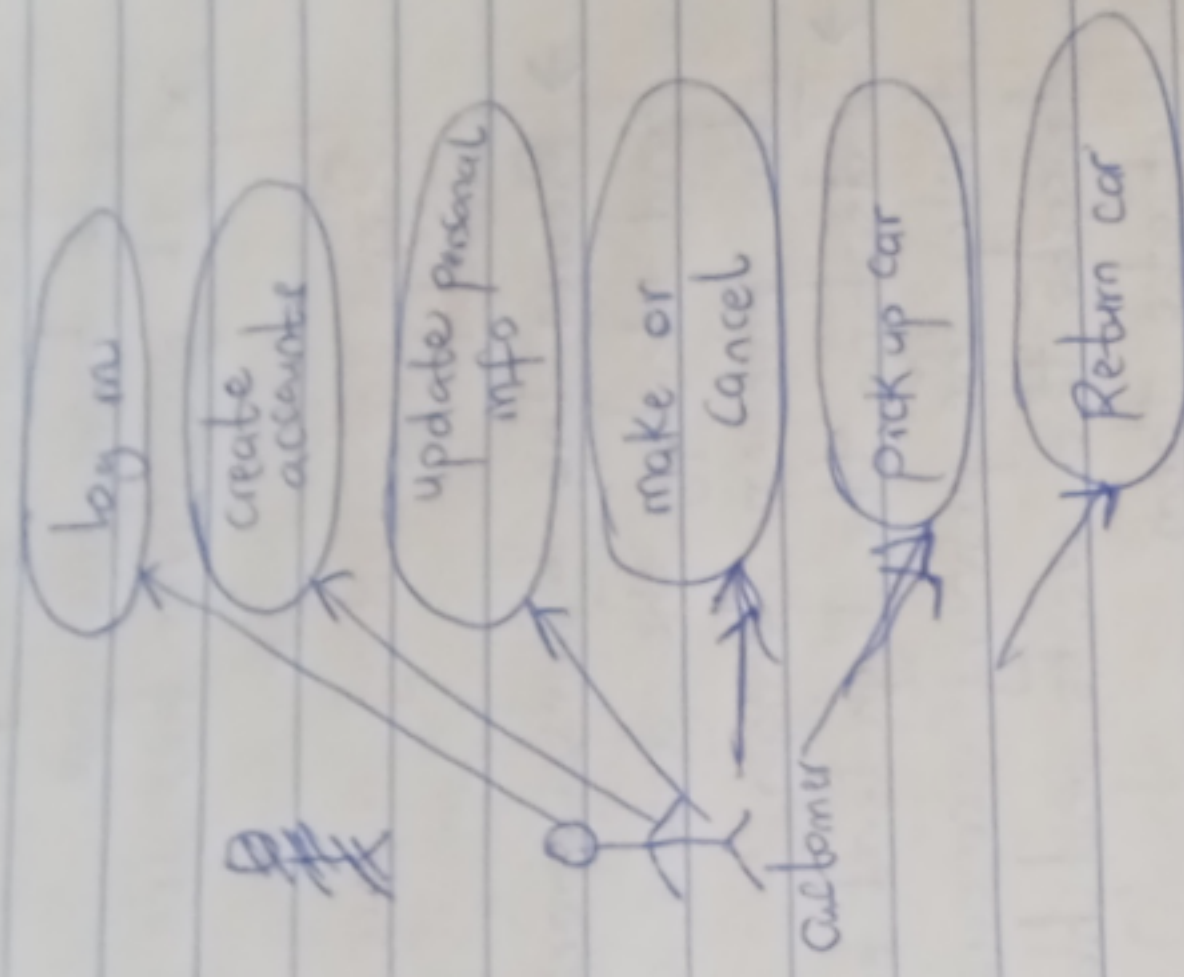


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Question

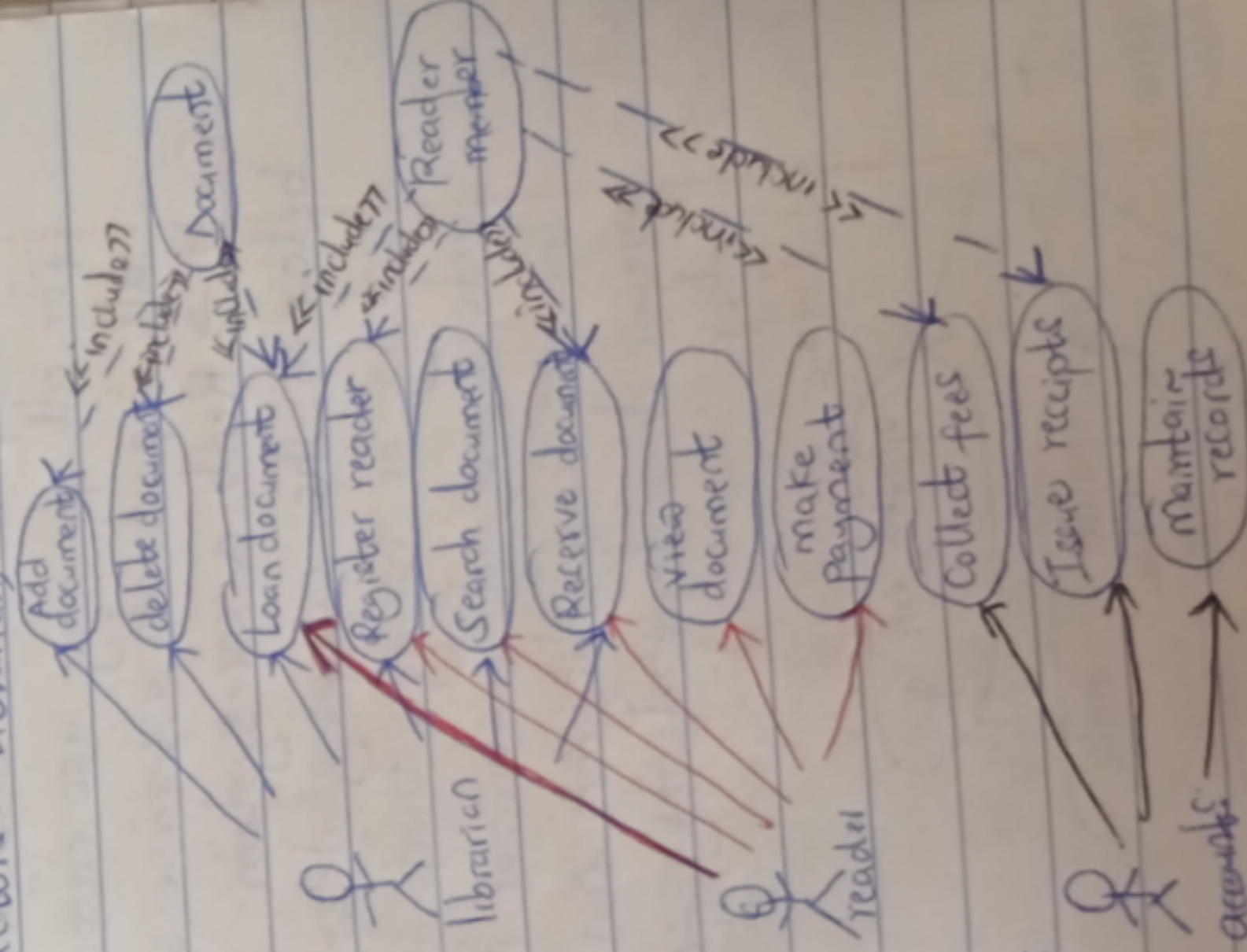
Consider developing a campus classroom facilities management system, the campus has a collection of classrooms. Each classroom has a number of equipments i.e microphone, whiteboard, overhead projector, tv, internet connection. Each equipment has info i.e history, condition, available forms as well as info specific to the equipment i.e size of whiteboard, t.v, mobile or not for microphone & speed of connection to the internet. Each classroom has a list of courses scheduled for the classroom, the instructor for each course requested equipments & setup for each course. A manager in charge of getting up the equipments. The system has to be developed which should allow instructors to book up the list of equipments in a classroom, request equipment and setup and query the setup status. The system should send

Customer Use case:



Ex. 2

This example is of a library management system, there are 3 actors: Librarian, accounts & reader



The domain problem is an ^{info} system for a video rental store simplifying assumptions & details:

1. It is a standalone shop not part of a larger organization.
2. Rents only videos, not computer games or other items.
3. A video can be in any media ie tape, dvd & so on.
4. the rental charge may vary by medium ie dvd rentals are more expensive than tapes.
5. The store does not sell anything it just rents.
6. All transactions ^{→ doing transactions} are rentals.
7. The input medium by which membership & video rentals are captured isn't important.
8. Cash only payments.
9. On completion of a rental, the customer receives a transaction report with typical info on the use of his/her judgement.

Identify at least all ^{four} primary use cases & related primary actors, add the attributes ie extend & include,

request for equipment and setups in a classroom to the classroom manager. The manager can update equipment info and set up status, ^{increase} of an equipment cannot be set up according to the request, the manager should notify the faculty as soon as possible. The system should also allow ^{anyone} ~~someone~~ to send comments to the manager. Develop UML model of the system using use case diagrams

b) class diagrams
c) statechart diagrams
d) Sequence diagrams
and have implement include & extend property in use case

